

MD RAKIBUL ISLAM RAIHAN

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Xi'an, China | Bangladesh | English (Fluent, IELTS 8.0) | Chinese (Conversational, HSK 3)

SUMMARY

M.Eng. Software Engineering student (Top 1%) specializing in LLM inference systems and GPU infrastructure. Experienced in designing and optimizing distributed model serving pipelines using SGLang, CUDA, and Triton with FP8/FP4 quantization across multi-GPU clusters (A800/H200). Achieved 8.3× reduction in time-to-first-token and 92% KV-cache hit rate in production-like systems. Researcher in multimodal AI (NVFP4-DiT, IEEE TMM under review). Seeking AI Infrastructure Internship (Summer 2026) with full-time conversion (2027).

SKILLS

- **LLM Systems & Serving:** SGLang, vLLM, TensorRT-LLM, Transformer Engine, RDMA (Mooncake)
- **GPU & Performance Engineering:** CUDA, Triton, FP8/FP4 quantization, GEMM optimization, Nsight Systems/Compute
- **Distributed Systems:** KV caching, PD disaggregation, multi-GPU inference, Linux
- **ML Frameworks:** PyTorch, Transformers
- **Languages:** Python (Expert), C++, Java, SQL

EXPERIENCE

AI Infrastructure Engineer Intern | InfiX.ai | Shenzhen, China | Apr 2026 – Jun 2026

- Optimized end-to-end LLM inference pipelines across A800/H200 GPU clusters using SGLang + Mooncake RDMA (Qwen, DeepSeek models)
- Designed hierarchical KV-cache system achieving 92% hit rate and 8.3× TTFT reduction (55s → 6.7s) across 4M+ tokens
- Built PD-disaggregated execution engine enabling dynamic prefill/decode scheduling across heterogeneous GPUs
- Developed CUDA/Triton kernels for FP8/FP4 compute and eliminated GEMM bottlenecks via profiling-driven optimization
- Analyzed cache policies (LRU/LFU/SLRU), identifying LFU as optimal under high-load inference scenarios; resolved production issues

Software Engineer Intern (AI Agent) | Hong Kong Amraim Group | Shenzhen, China | Jun 2025 – Sep 2025

- Built production full-stack systems using React and REST APIs
- Developed AI agents for workflow automation, improving operational efficiency by 25%
- Delivered production-ready services using Agile development practices

Electrical Software Engineer Intern | Shaanxi Longong Intelligent Technology | Xi'an, China | Feb 2025 – Apr 2025

- Developed PLC-based industrial control systems (CODESYS) for robotics automation
- Implemented Modbus, CAN, and Ethernet/IP communication protocols
- Improved motor control performance by 15% through system-level optimization

RESEARCH EXPERIENCE

Graduate Researcher | Northwestern Polytechnical University | Xi'an, China | Sep 2024 – Mar 2027

- Developed NVFP4-DiT: 4-bit multimodal diffusion transformer for audio-guided video generation
- Achieved 4× memory reduction while maintaining generation quality
- Paper under review: "NVFP4-DiT: Efficient 4-Bit Audio-Guided Video Diffusion Transformers for Low-Cost Video Generation" (IEEE Transactions on Multimedia, Q1)
- M.Eng. Thesis: "Detecting Deepfake Videos: A Multimodal AI Framework Leveraging Visual, Auditory, and Temporal Inconsistencies" (Advisor: Dr. Zhang Peng)

Undergraduate Researcher | Northwestern Polytechnical University | Xi'an, China | Jan 2023 – Jul 2024

- Designed distributed confidential query system using Apache Spark with homomorphic encryption
- Achieved 95% privacy preservation with 20% performance improvement
- B.Eng. Thesis: "Design and Implementation of a Distributed Confidential Query Protocol for Apache Spark" (Advisor: Dr. Chen Yaxing)

EDUCATION

M.Eng. Software Engineering | Northwestern Polytechnical University (985/211) | Sep 2024 – Mar 2027

GPA: 88/100 | Top 1%

B.Eng. Computer Science and Technology | Northwestern Polytechnical University (985/211) | Sep 2020 – Jul 2024

GPA: 85/100 | Top 1%

AWARDS & CERTIFICATIONS

Chinese Government Scholarship (2024–2026) | NWPU Presidential Scholarship (2020–2023) | Sanqin Scholarship (2023) | Wu Yajun Scholarship (2024) | Director, International Students Union (2023) | Hackathon Winner – NWPU AI Challenge (2022) | Best Undergraduate Thesis Award | Outstanding Graduate (NWPU, 2024)

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中国西安 | 孟加拉国 | 英语 (流利, 雅思 8.0) | 中文 (日常交流, HSK 3)

个人摘要

软件工程硕士研究生 (专业前 1%), 专注于大语言模型推理系统与 GPU 基础设施。具备使用 SGLang、CUDA 和 Triton 在多 GPU 集群 (A800/H200) 上设计和优化分布式模型服务流水线的经验, 并应用 FP8/FP4 量化技术。实现了首令牌时延 8.3 倍降低和 92% KV 缓存命中率。多模态 AI 研究员 (NVFP4-DiT, IEEE TMM 在审)。寻求 AI 基础设施实习 (2026 年暑期), 并期望在 2027 年毕业后转为全职。

专业技能

- LLM 系统与服务: SGLang, vLLM, TensorRT-LLM, Transformer Engine, RDMA (Mooncake)
- GPU 与性能工程: CUDA, Triton, FP8/FP4 量化, GEMM 优化, Nsight Systems/Compute
- 分布式系统: KV 缓存, PD 分离推理, 多 GPU 推理, Linux
- ML 框架: PyTorch, Transformers
- 编程语言: Python (精通), C++, Java, SQL

实习经历

AI 基础设施工程师实习生 | 无界智索(深圳)科技有限责任公司 | 中国深圳 | 2026 年 4 月 – 2026 年 6 月

- 使用 SGLang 和 Mooncake RDMA 在 A800/H200 GPU 集群上优化端到端 LLM 推理流水线 (Qwen、DeepSeek 等模型)
- 设计分层 KV 缓存系统, 实现 92% 缓存命中率和 8.3 倍首令牌时延降低 (55 秒 → 6.7 秒), 处理 400 万+ tokens
- 构建 PD 分离推理执行引擎, 实现跨异构 GPU 的动态预填充/解码调度
- 开发 CUDA/Triton 内核以支持 FP8/FP4 计算, 并通过性能分析驱动优化消除 GEMM 瓶颈
- 分析缓存策略 (LRU/LFU/SLRU), 证明 LFU 在高负载推理场景下最优; 解决生产环境中的推理问题

人工智能工程师实习生 (AI 代理) | 香港安莱姆集团有限公司 | 中国深圳 | 2025 年 6 月 – 2025 年 9 月

- 使用 React 和 REST API 构建生产级全栈系统
- 设计 AI 智能体实现工作流自动化, 运营效率提升 25%
- 采用敏捷开发实践交付生产级服务

电气软件工程师实习生 | 陕西龙宫智能科技有限公司 | 中国西安 | 2025 年 2 月 – 2025 年 4 月

- 使用 CODESYS 为机器人自动化开发基于 PLC 的工业控制系统
- 实现 Modbus、CAN 和 Ethernet/IP 通信协议
- 通过系统级优化将电机控制性能提升 15%

研究经历

研究生研究员 | 西北工业大学 | 中国西安 | 2024 年 9 月 – 2027 年 3 月

- 开发 NVFP4-DiT: 用于音频引导视频生成的 4 位多模态扩散 Transformer
- 实现 4 倍内存压缩, 同时保持生成质量
- 论文在审: "NVFP4-DiT: Efficient 4-Bit Audio-Guided Video Diffusion Transformers for Low-Cost Video Generation" (IEEE Transactions on Multimedia, Q1)
- 硕士论文: "Detecting Deepfake Videos: A Multimodal AI Framework Leveraging Visual, Auditory, and Temporal Inconsistencies" (导师: 张蓬博士)

本科生研究员 | 西北工业大学 | 中国西安 | 2023 年 1 月 – 2024 年 7 月

- 使用 Apache Spark 和同态加密设计分布式机密查询系统
- 实现 95% 隐私保护, 性能较基准提升 20%
- 本科论文: "Design and Implementation of a Distributed Confidential Query Protocol for Apache Spark" (导师: 陈亚兴博士)

教育背景

软件工程硕士 | 西北工业大学 (985/211) | 2024 年 9 月 – 2027 年 3 月

GPA: 88/100 | 专业前 1%

计算机科学与技术学士 | 西北工业大学 (985/211) | 2020 年 9 月 – 2024 年 7 月

GPA: 85/100 | 专业前 1%

奖项与证书

中国政府奖学金 (2024–2026) | 西北工业大学校长奖学金 (2020–2023) | 三秦奖学金 (2023) | 吴亚军奖学金 (2024) | 国际学生会主席 (2023) | 西北工业大学 AI 挑战赛黑客松冠军 (2022) | 本科优秀毕业论文奖 | 优秀毕业生 (西北工业大学, 2024)